Target shielding in a PET-production facility: Effect on the activation of the biological shield?

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VUB-UZ Brussel class IIA facility

- Located inside UZ Brussel, Laarbeeklaan 103, 1090 Jette



- CGR560 cyclotron: operation 1985-2020, currently under decommissioning
- IBA Cyclone® KIUBE: commissioned end 2017
- GMP-radiopharmacy

VUB-UZ Brussel class IIA facility

CGR 560 + irradiation room facilities

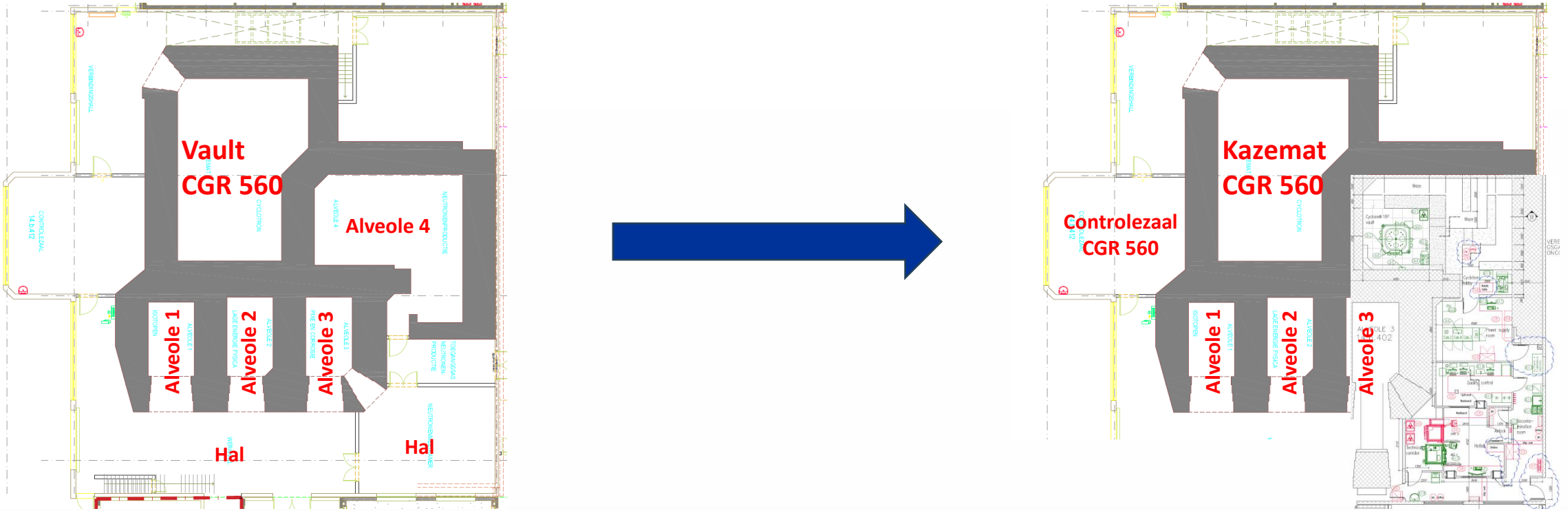


IBA Cyclone® KIUBE



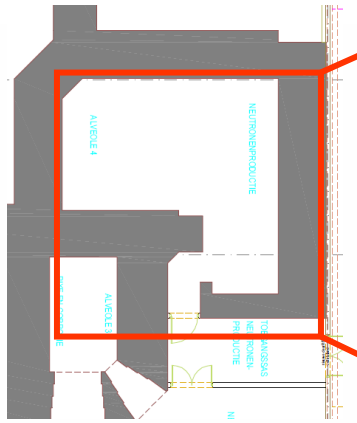
Some history

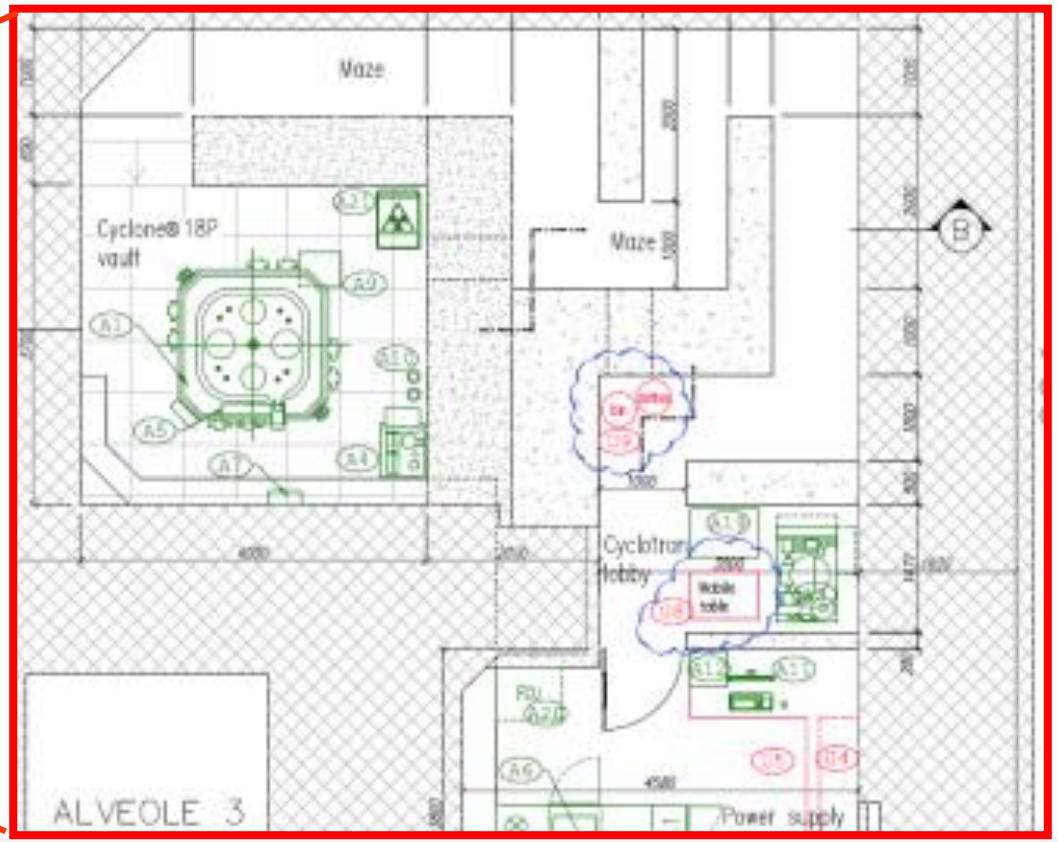
- 2016: transformation of CGR-irradiation room 4 into IBA Cyclone® KIUBE vault



Some history

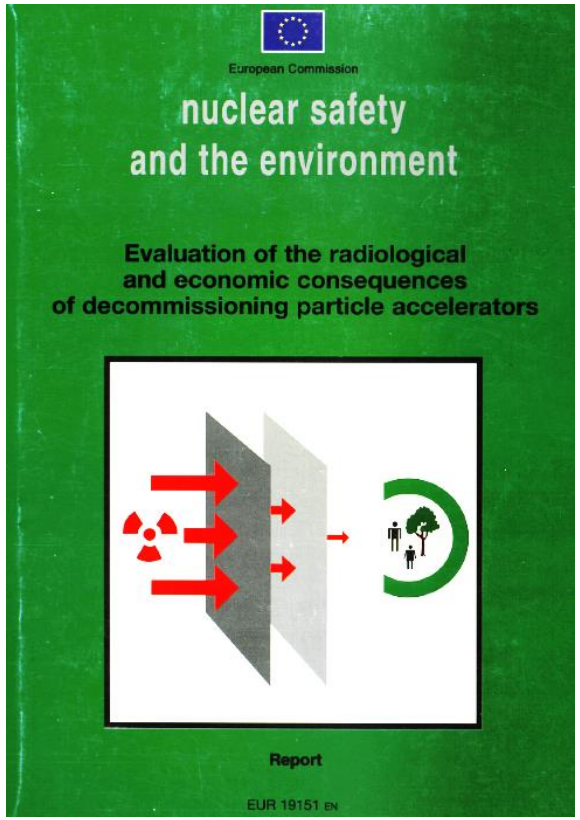
- 2016: transformation of CGR-irradiation room 4 into IBA Cyclone® KIUBE vault

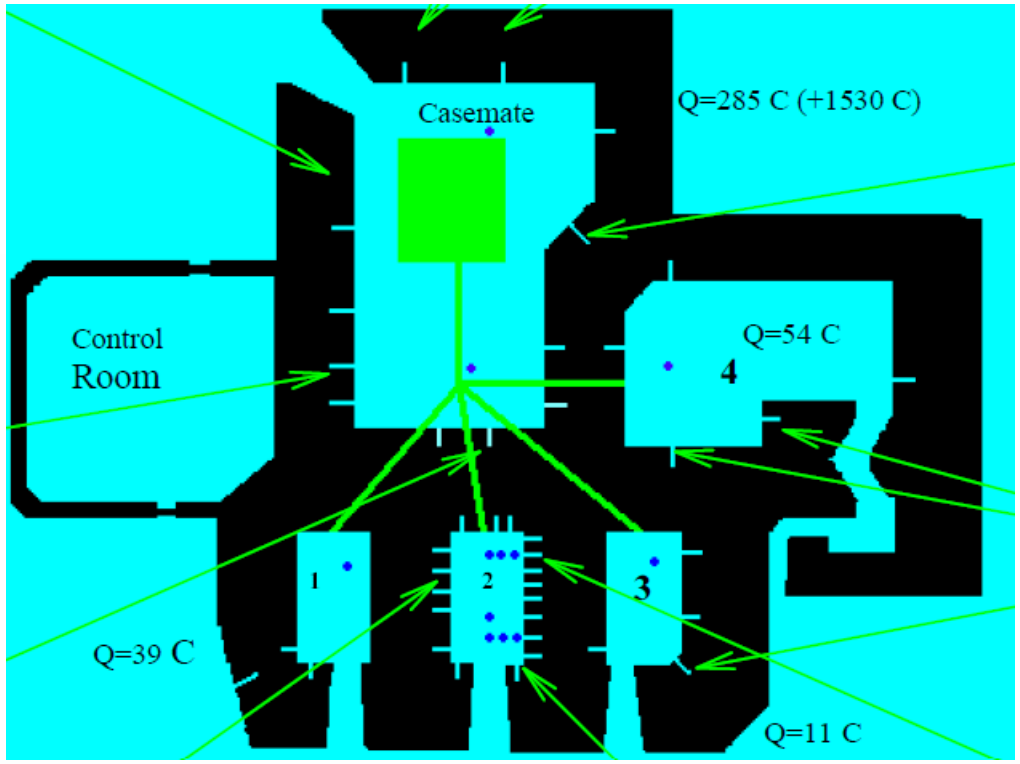




Some history

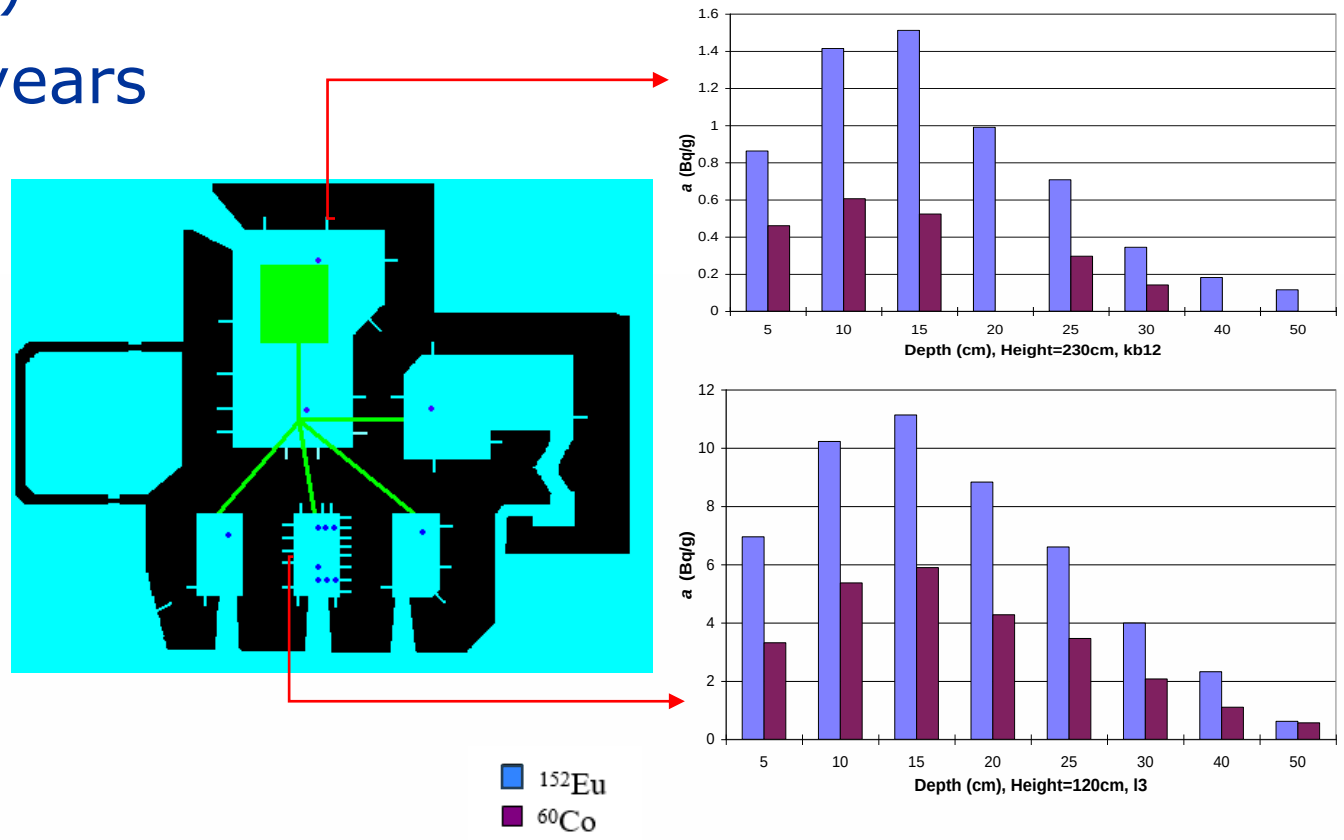
- What we already knew (1999)





Some history

- What we already knew (1999)
- Concrete activation after 10 years of operation
- CGR vault
  - ^{152}EU: 1.5 Bq/g
  - ^{60}Co: 0.6 Bq/g
- Alveole 2
  - ^{152}EU: 13 Bq/g
  - ^{60}Co: 6 Bq/g



Some history

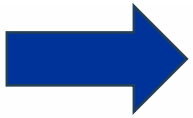
- Cause of activation
  - (p,xn) reactions in CGR-cyclotron
  - Alveole 2: production of ^{201}Tl

    $^{203}Tl(p,3n)^{201}Pb \rightarrow ^{201}Tl$

  - Relatively high cross-sections for thermal neutron activation

    $^{151}Eu(n,\gamma)^{152}Eu$: 9198 barn

    $^{59}Co(n,\gamma)^{60}Co$: 37 barn

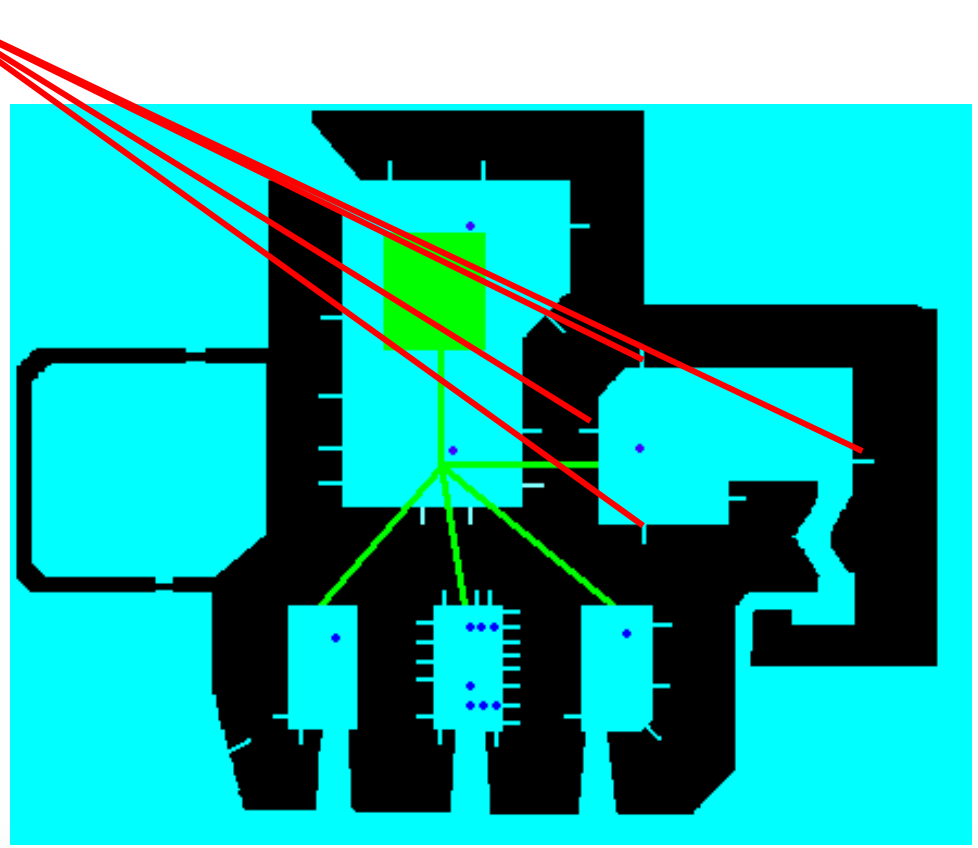


> 70m^3 activated concrete
> unconditional clearance levels ($^{152}EU/^{60}Co$)

Installation of the IBA Cyclone® KIUBE

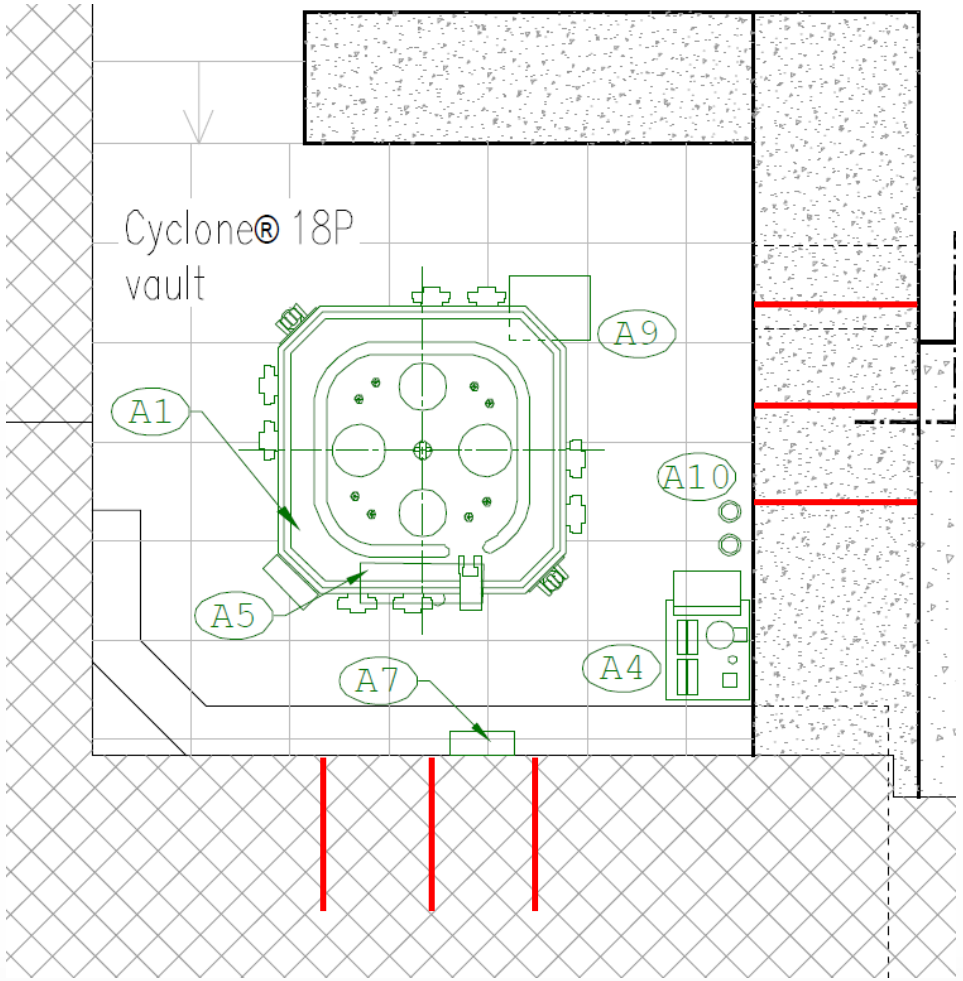
- Alveole 4: activation levels < clearance levels

Drillings performed before the installation to monitor future activation



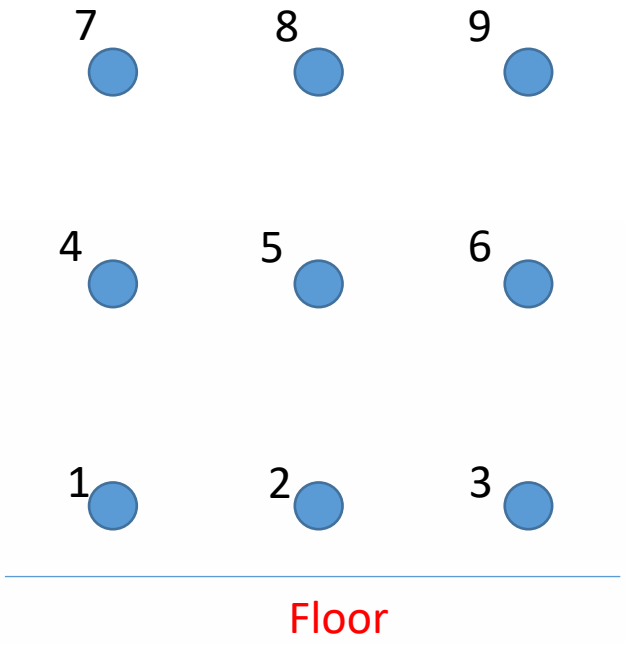
Drillings IBA Cyclone® KIUBE Vault

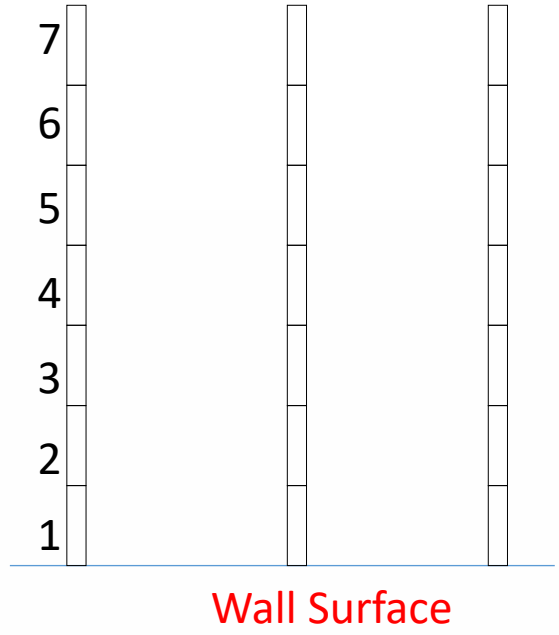
- Two walls
- Each wall 9 locations in a grid of 3x3
- Total depth: ± 100 cm
- 7 rods for each location (∅ 4 cm, length ± 150 cm)
- Easy to remove as rods can be put back in a (blind) PVC-tube



Drillings IBA Cyclone® KIUBE Vault

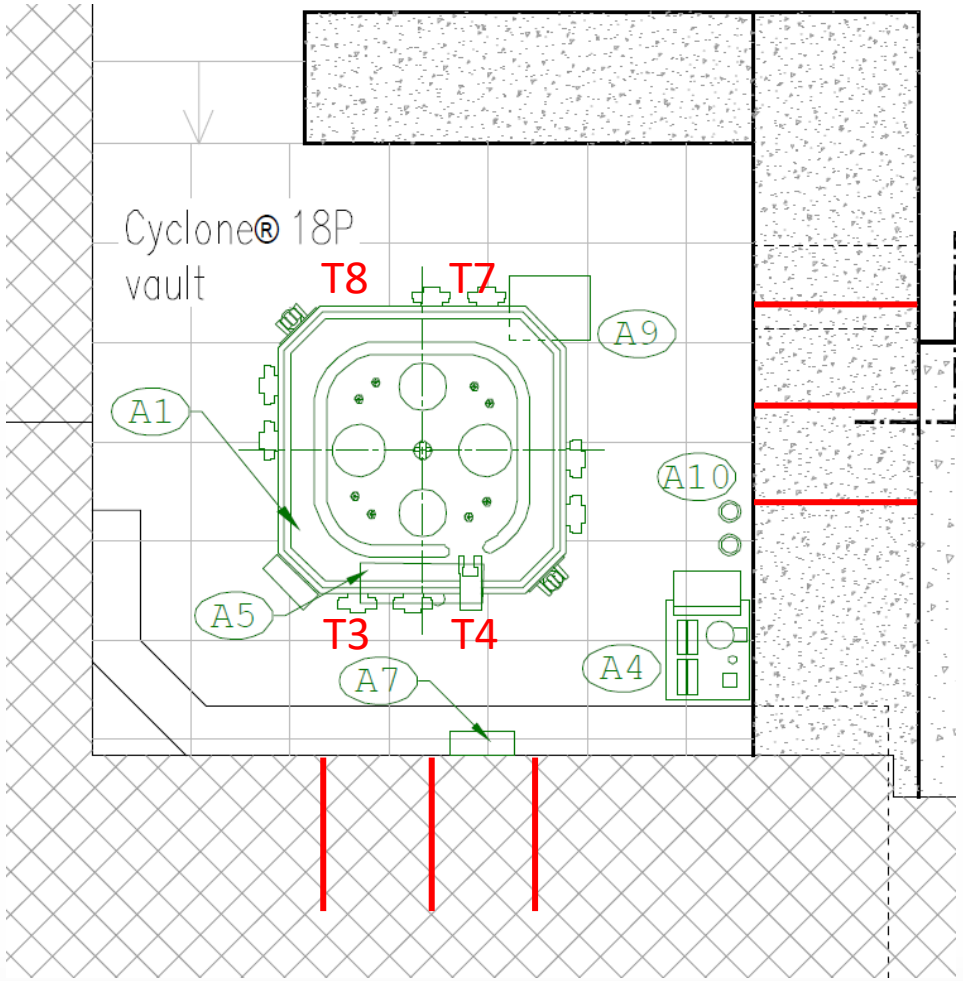
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Target location IBA Cyclone® KIUBE Vault

- 8 ports on Cyclone® KIUBE
- 4 out 8 ports used for Nirta® 18F conical liquid target
- 2 x 2XL on port 3 and 7
- 2 x 3XL on port 4 and 8



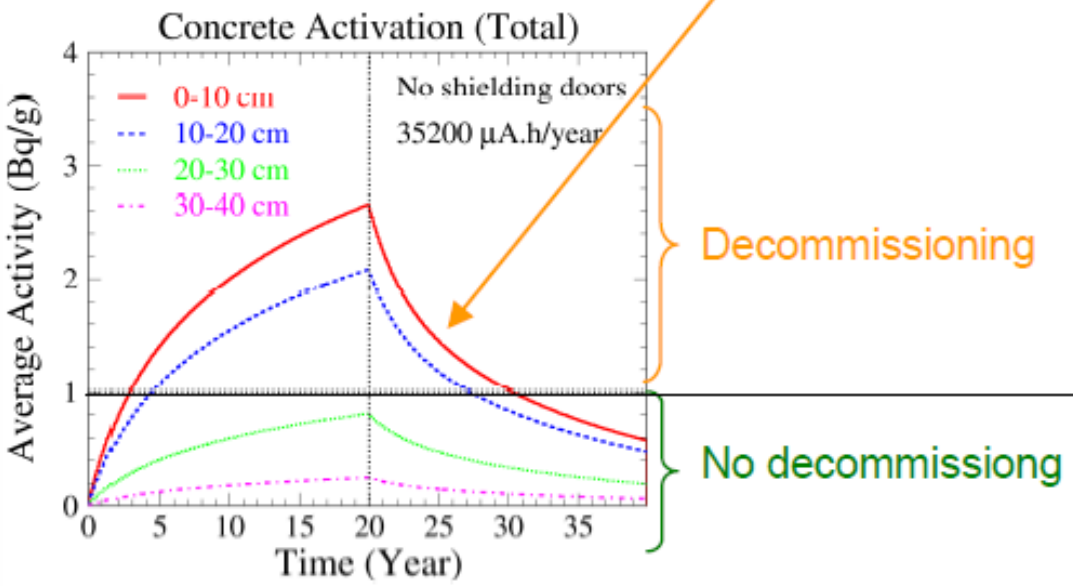
Installation Nirta® shielded doors

- On each target
- Neutron absorber (HDPE)
- Limit air activation
- Limit activation accelerator parts
- Limit activation of biological shield
- 30kEUR investment



Technical datasheet Nirta® shielded door

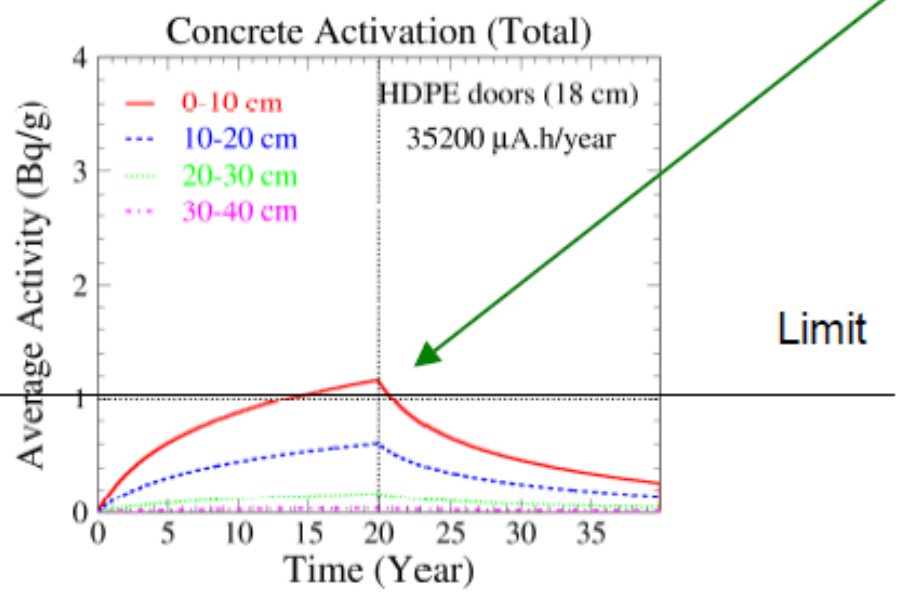
Without local shielded doors, a layer of the vault walls is considered as radioactive waste



Decommissioning

No decommissiong

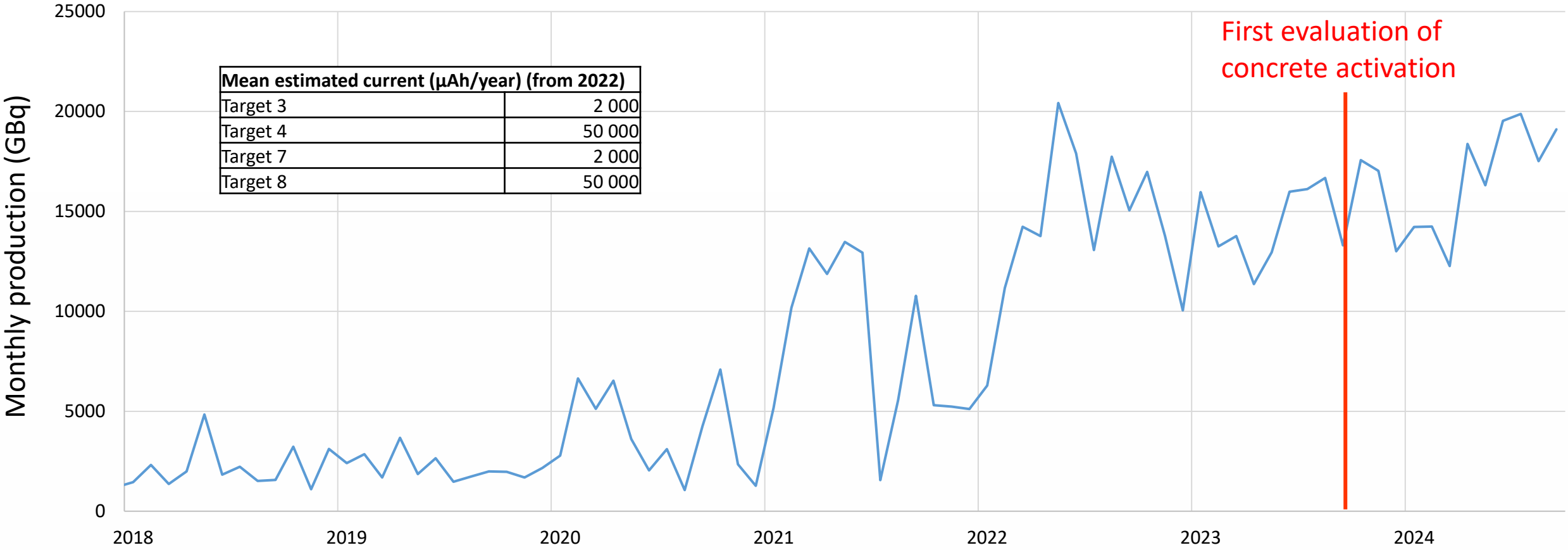
With local shielded doors, after one year without operation, vault walls activation is below the limit.



Limit ?

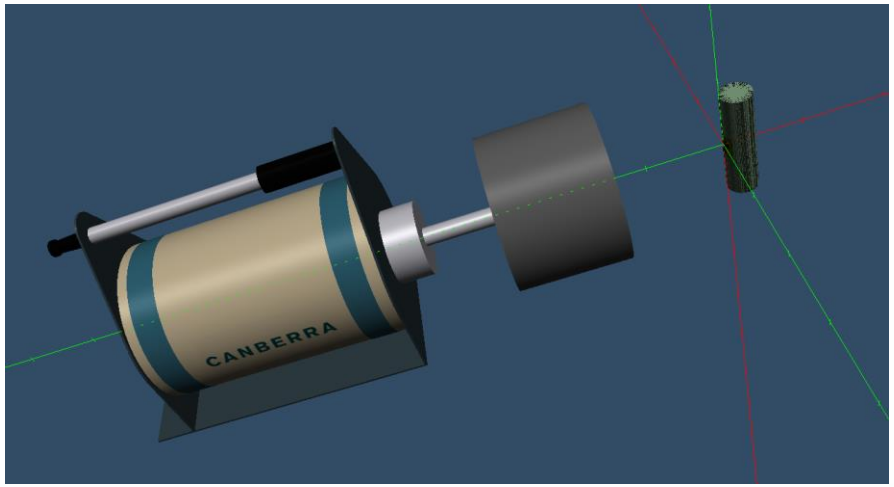
VUB-UZ workload Cyclone® KIUBE Vault

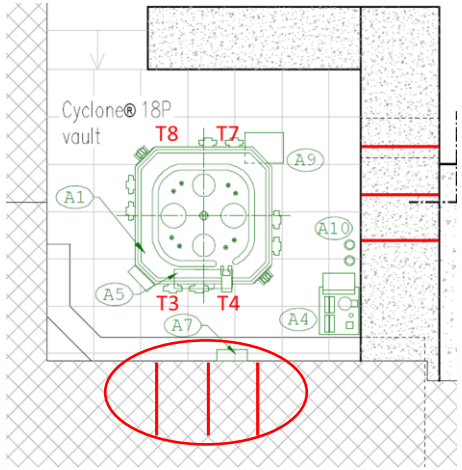
Mean estimated current (µAh/year) (from 2022)	
Target 3	2 000
Target 4	50 000
Target 7	2 000
Target 8	50 000

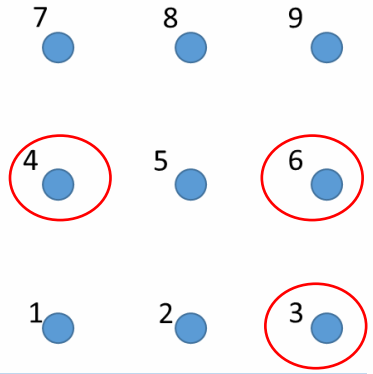


First analysis of concrete rods

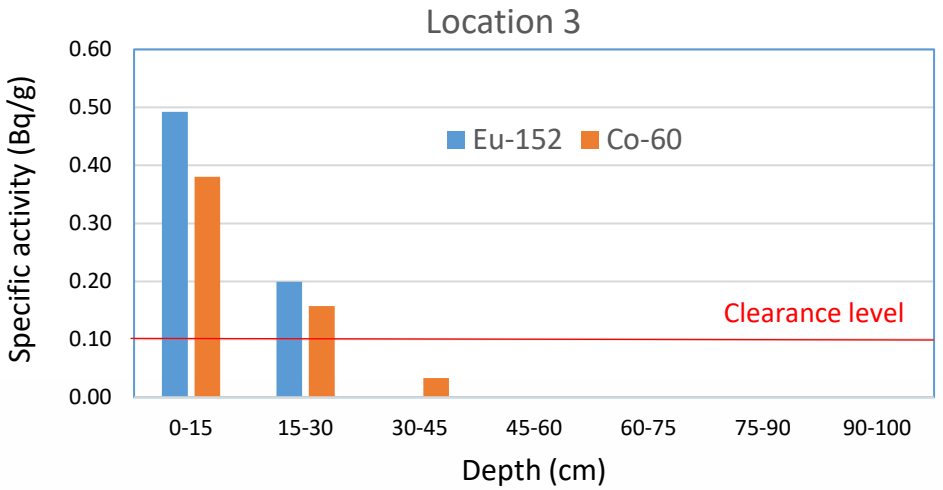
- Three locations in front of target 3 and target 4 (location 3, 4 and 6)
- Characterisation of rods with ISOCS™
- MDA's: 0.03-0.06 Bq/g (2h counting time)

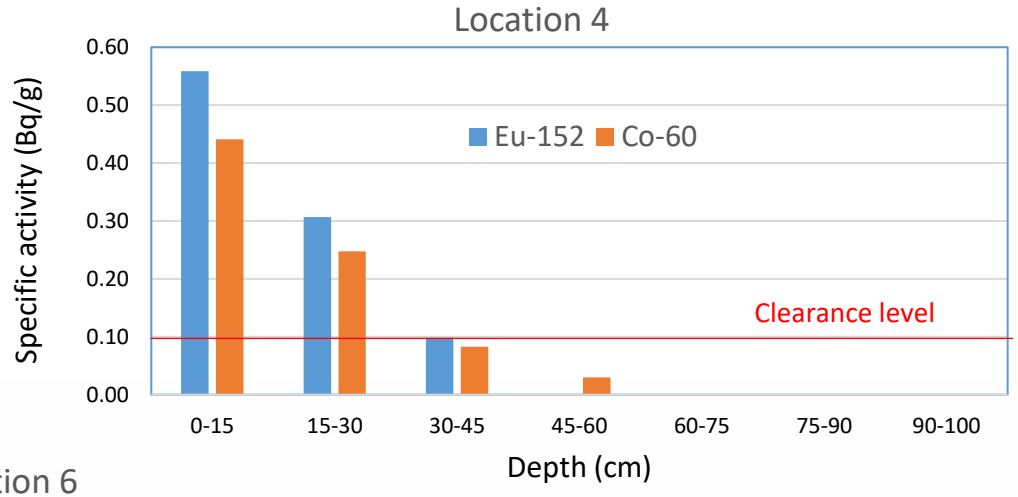


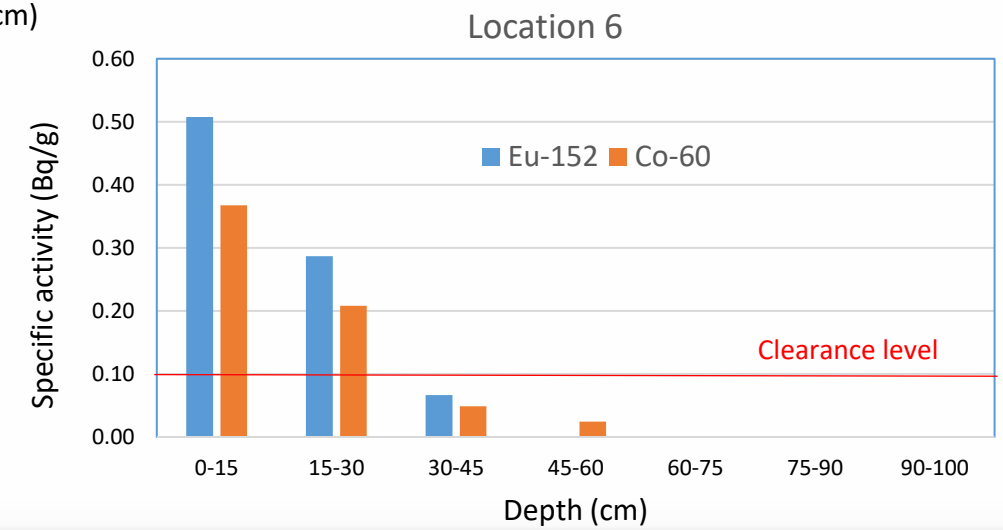


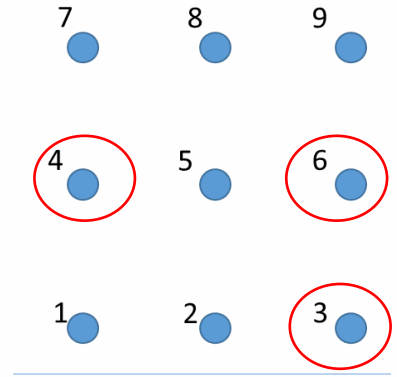


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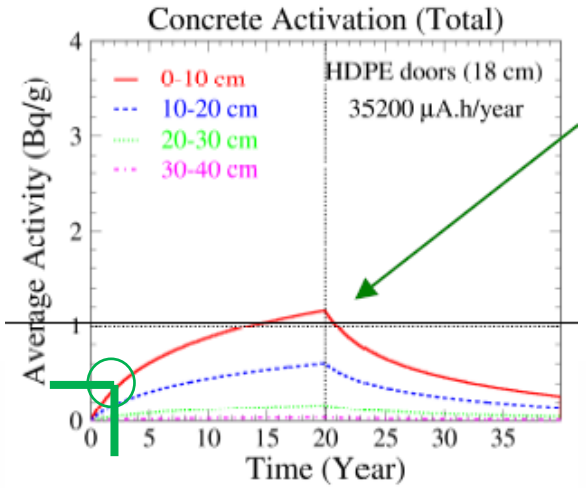






Efficacy of shielded doors

- Measurement results in line with predicted data



- Activation > clearance levels after two years of routine production
- Impact on other walls to be investigated

Some discussion points

- Cost-effectiveness after reducing activation level by a factor of 2-3?
- Activation > clearance level
  Real impact on future decommissioning strategies?
- Lessons for new installations?